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Feature Selection in Censored-Covariate Regression Models

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ABSTRACT

Statistical methods are increasingly needed to identify important features in regression models fitted on incomplete biomedical data, in order to help improve scientific understanding and inform clinical decision-making. One important example of incomplete data is censored covariates—some biomedical studies have an important covariate that is known to affect the response variable and must be included in the regression model, but this covariate may be only partially observed, for example due to a time-to-event right-censoring mechanism. Feature-selection methods must account for the unobserved part of that covariate; however, existing methods for feature selection typically focus on when the response variable is censored rather than when a covariate is censored. We propose a penalised observed-data log-likelihood method for doing feature selection in censored-covariate regression models, where we use two separate L_1 penalties to select two sets of features: one set for the response-variable model component and one set for the censored-covariate model component. We derive an Expectation-Minimisation (EM) algorithm for our method. A simulation study shows that the method recovers the active feature sets when the two model components share all, some, or none of these features. Applied to PREDICT-HD data, we show that the method selects partly distinct brain-region profiles for cognitive ability and time-to-diagnosis of Huntington disease. Researchers applying censored-covariate regression models to high-dimensional data sets can use our method to identify important features associated with the response variable and how they may differ to those associated with the censored covariate.

1 | Introduction

1.1 | Background

With the increase in biomedical data from electronic health records, imaging studies, and clinical trials, statistical methods are increasingly needed to identify important features from incomplete data. Identifying these features can improve scientific understanding, prediction, and clinical decision-making. However, incomplete data can otherwise lead to biased conclusions or missed associations if the incomplete-data mechanism is ignored.

Censored covariates in regression modelling are one important example of this incomplete-data problem. They arise when a response variable is modelled using covariates that are only partially observed due to censoring; in this paper, we focus on the case where there is only one censored covariate. In biomedical

studies, censoring in covariates can occur when biomarker concentrations fall below laboratory detection limits [1], or when a time-to-event quantity is used as a covariate but the event has not been observed for all individuals by the end of follow-up [2, 3], among other possible reasons [4, 5, 6]. Naive methods for fitting censored-covariate regression models, such as complete-case analysis or replacing censored values by fixed constants, can lead to biased or inefficient estimation because they ignore the latent part of the covariate distribution. Other methods, such as minimising an observed-data likelihood that integrates out the censored covariate by assuming it follows a specific distribution, have therefore been developed to account for censored covariates; a review of these methods can be found in Lotspeich et al. [7].

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Feature selection for censored-covariate regression models requires methods that account for the fact that the covariate entering the regression model is only partially observed. We use the term ‘feature’ in this paper to refer to any fully-observed covariate that also enters the regression model, meaning that only fully-observed covariates are prone to being excluded under feature selection. Feature selection is well developed for complete data—penalised regression methods such as the lasso [8] provide a way to do feature selection by shrinking coefficients exactly to zero—and has also been widely studied when the response variable is censored, including methods for survival [9, 10, 11, 12], interval-censored [13], multistate [14], and related models. Feature selection has also been considered in joint longitudinal and time-to-event models [15], a modelling framework designed to describe the association between a longitudinal process and a time-to-event process [16, 17, 18, 19, 20], and in regression models with missing data [21, 22, 23, 24]. None of the mentioned feature-selection methods can directly be applied when the covariate is censored rather than the response variable because the likelihood of the data needs to account for the covariate-censoring mechanism by integrating out the covariate. To the best of our knowledge, existing censored-covariate regression methods focus primarily on parameter estimation and inference for pre-specified models, rather than on feature selection to obtain a sparser model. Consequently, researchers cannot identify features associated with the response variable while accounting for the covariate-censoring mechanism. In accounting for this censoring mechanism, researchers may also be interested in identifying features associated with the censored covariate, which is also not possible without a tailored feature-selection method. A key example of where features need to be identified not just for the response variable but also for the censored covariate, is in Huntington disease applications (Section 1.2), where time until diagnosis of the disease may be right-censored but remains scientifically important. This gap matters because researchers must otherwise rely on ad hoc methods or pre-specified models, leading to less reliable scientific conclusions.

Existing censored-covariate regression methods therefore leave two related feature-selection questions unresolved: which fully-observed features are associated with the response after accounting for the censored covariate, and which are associated with the censored covariate itself. We address both questions by developing a feature-selection method for regression models with a single right-censored covariate, where we introduce two separate penalties to select a set of features associated with the response variable and a set of features associated with the censored covariate, allowing the two sets to overlap, differ, or be disjoint. This distinction is important because features that help explain the response variable need not be the same as features associated with the censored covariate. Unlike existing feature-selection methods for censored data, which typically handle censoring in the response variable, our method is derived for the setting where the covariate is right-censored and enables feature selection for both parts of the model. Although we focus on a right-censored covariate, the method could be adapted to other types of censoring as well.

1.2 | Huntington Disease Data

Huntington disease (HD) is a neurodegenerative disease caused by a mutated Huntingtin gene. Carrying the mutated gene is hereditary, so it is known from birth whether an individual is guaranteed to eventually be diagnosed with HD. Clinicians are interested in understanding what drives HD progression because HD has no cure, and understanding these drivers is important for prognosis and for designing clinical trials, which are essential for evaluating potential therapies.

The PREDICT-HD study was a large observational study from 2001 to 2013 that recruited individuals who carried the mutated Huntingtin gene but were not yet diagnosed with HD, and tracked several of their measurements over time. Research that analysed data from the PREDICT-HD study found that there was a relationship between an individual’s cognitive ability and their estimated time-to-diagnosis (TTD) of HD [25]. These two variables are distinct but related measures of HD progression.

Volumes of certain brain regions are associated with both cognitive ability and TTD because HD is characterised by subcortical atrophy, meaning shrinkage of brain structures located below the cerebral cortex. One such subcortical structure is the striatum, a group of brain regions including the caudate and putamen, which are involved in movement and cognition. The caudate and putamen can see volume loss years before diagnosis, with further volume loss leading up to diagnosis [26]. However, it remains unclear whether other brain regions are also associated with time-to-diagnosis, or with cognitive ability, or with both.

Our aim is therefore to select potentially different sets of brain regions associated with cognitive ability—measured by the Symbol Digit Modalities Test (SDMT) score—and TTD, whilst accounting for the relationship between these two variables as identified by previous research. Diagnosis of HD is typically done using the Diagnostic Confidence Level scale, which is based on an individual’s motor ability rather than their cognitive ability. Hence, it is entirely possible that different brain regions may be associated with TTD compared to those with SDMT score, which measures cognitive ability. Identifying whether these two variables are associated with shared or distinct brain-region profiles is important for prognosis and for designing or enriching clinical trials, especially because SDMT score is a candidate cognitive endpoint in HD studies [27].

The method developed in this paper is able to address our aim because it selects features for both the response variable and the censored covariate. To apply the method, we use PREDICT-HD measurements at study entry, with SDMT score as the response variable, TTD as the censored covariate, and subcortical brain-region volumes as features. We treat SDMT score as the response because it is the cognitive outcome of interest, and we treat TTD as a covariate because previous research indicates that cognitive ability is related to how close an individual is to HD diagnosis. TTD is right-censored for individuals who were never diagnosed with HD during the study, so this application matches the censored-covariate regression setting. The full data set construction and application of our method are described in Section 4.1.

The remainder of the paper is organised as follows. Section 2 presents the censored-covariate regression model, explains how

we perform feature selection by adding L_1 penalties, describes parameter estimation using an Expectation-Minimisation (EM) algorithm, and explains how tuning parameters are chosen. Section 3 evaluates our method on selection performance and parameter estimation error in a simulation study, and Section 4 applies the method to the Huntington disease data. Finally, Section 5 discusses the main findings, potential impact, limitations, and directions for future work.

2 | Method

We propose a method to do feature selection in censored-covariate regression models. Our method jointly models the response variable y and the censored covariate x to obtain an observed-data log-likelihood, and then augments this log-likelihood with separate L_1 penalties on two coefficient vectors β and γ , resulting in a penalised observed-data loss function. Non-zero entries in the estimated coefficient vector $\hat{\beta}$ correspond to features associated with y , while non-zero entries in the estimated coefficient vector $\hat{\gamma}$ correspond to features associated with x . Our method is therefore able to select potentially different sets of features for the response-variable model component and the censored-covariate model component of a censored-covariate regression model.

In this section, we first introduce the censored-covariate regression model and derive the penalised observed-data loss function, then describe parameter estimation via an Expectation-Minimisation (EM) algorithm, and finally discuss how to choose the two tuning parameters, λ_1 and λ_2 , that control the strengths of the two L_1 penalties.

2.1 | Model

We consider the censored-covariate regression model given by

$$y_i = \beta_0 + \alpha x_i + \beta^\top \mathbf{z}_i + \varepsilon_i, \quad (1)$$

where y_i is the response variable, $x_i > 0$ is the single right-censored covariate with coefficient α , $\mathbf{z}_i$ is the p -dimensional vector of features with coefficient vector $\beta = [\beta_1, \dots, \beta_p]$, β_0 is the intercept, and $\varepsilon_i \sim N(0, \sigma^2)$. The subscript $i = 1, \dots, n$ indexes the observations in the data set, and we assume that observations are independent of each other. The covariate x_i is right-censored by a censoring variable c_i , so that if $c_i < x_i$, then c_i is observed instead of x_i ; we denote the censoring indicator by $\delta_i = 1_{\{x_i \leq c_i\}}$, and the value of δ_i is known for each observation. We refer to observations with $\delta_i = 0$ ($\delta_i = 1$) as observations with censored x_i (complete x_i).

Unlike standard censored-covariate regression methods [7] that treat x_i as a nuisance, we regard x_i as a variable of scientific interest much like y_i , so we wish to understand which features are associated with x_i . Letting uppercase letters denote the random-variable versions of lowercase letters, we model the distribution of X_i conditional on $\mathbf{Z}_i = \mathbf{z}_i$ as

$$X_i | \mathbf{Z}_i = \mathbf{z}_i \sim \chi^2(\mu_i), \quad (2)$$

where the mean is given by $\mu_i = \exp(\gamma_0 + \gamma^\top \mathbf{z}_i)$. Here, γ_0 is an intercept and $\gamma = [\gamma_1, \dots, \gamma_p]$ is a coefficient vector such that non-zero entries of γ correspond to features associated with x_i . We will refer to (1) and (2) together as the *censored-covariate regression model*, and refer to (1) as the *response-variable model component* and (2) as the *censored-covariate model component*.

Motivated by our application in Section 4 where x_i is an individual's time until diagnosis of a disease so that μ_i is the expected time until diagnosis for an individual with features $\mathbf{z}_i$, we choose to use a χ^2 distribution in (2). The χ^2 distribution is positively valued and much of its mass lies close to its mean as μ_i increases, which largely reflects the application: for a low-risk individual with large μ_i , we expect their realised x_i to fall near μ_i . Additionally, the χ^2 distribution is a one-parameter distribution, which, while may be restrictive, simplifies parameter estimation.

To estimate the parameters of (1) and (2) while accounting for censoring in X_i , we build up to an observed-data log-likelihood in stages. We begin by jointly modelling Y_i and X_i conditional on $\mathbf{Z}_i = \mathbf{z}_i$. Let $\theta = [\alpha, \beta, \beta_0, \sigma^2, \gamma, \gamma_0]$ denote the vector of parameters to estimate. We use square-bracket notation with entries separated by commas to indicate a column vector; if entries are separated by white space instead, it is a row vector. We also use notation such as $f(y_i | x_i, \mathbf{z}_i; \theta)$ to represent the conditional density $f_{Y_i|X_i, \mathbf{Z}_i}(\cdot; \theta)$ evaluated at $[y_i, x_i, \mathbf{z}_i]$. Under the conditional non-informative censoring assumptions

$$Y_i \perp C_i | X_i, \mathbf{Z}_i \quad \text{and} \quad X_i \perp C_i | \mathbf{Z}_i,$$

the complete-data likelihood factors as

$$L_c(\theta) \propto \prod_{i=1}^n f(y_i | x_i, \mathbf{z}_i; \theta) f(x_i | \mathbf{z}_i; \theta),$$

where the factor that would involve c_i is absorbed into the proportionality constant since it does not depend on θ . Therefore, the complete-data log-likelihood, up to additive constants that do not involve θ , is

$$\ell_c(\theta) = \sum_{i=1}^n \{\log f(y_i | x_i, \mathbf{z}_i; \theta) + \log f(x_i | \mathbf{z}_i; \theta)\},$$

where $f(y_i | x_i, \mathbf{z}_i; \theta)$ is the normal density function from (1) and $f(x_i | \mathbf{z}_i; \theta)$ is the χ^2 density function from (2).

Since x_i is censored in some observations, the observed-data log-likelihood is obtained from $\ell_c(\theta)$ by retaining the contributions from observations with complete x_i and letting observations with censored x_i contribute via integration over (c_i, ∞) , giving

$$\begin{aligned} \ell_o(\theta) = & \sum_{i: \delta_i=1} \{\log f(y_i | x_i, \mathbf{z}_i; \theta) + \log f(x_i | \mathbf{z}_i; \theta)\} \\ & + \sum_{i: \delta_i=0} \left\{ \log \int_{c_i}^{\infty} f(y_i | x, \mathbf{z}_i; \theta) f(x | \mathbf{z}_i; \theta) dx \right\}. \end{aligned}$$

Since we hold scientific interest in both y and x , we want to do feature selection on both model components. We therefore augment $\ell_o(\theta)$ with separate L_1 penalties on β and γ , which allows the method to select a potentially different set of features for each model component. Specifically, we define the penalised

observed-data loss function

$$\ell_o^{\text{pen}}(\theta) = -\frac{1}{n}\ell_o(\theta) + \frac{\lambda_1}{\sigma^2}\|\beta\|_1 + \lambda_2\|\gamma\|_1, \quad (3)$$

where λ_1 and λ_2 are tuning parameters that control the strengths of their respective penalties. The scaling of λ_1 by $1/\sigma^2$ is inspired by the ‘natural lasso’ of Yu and Bien [28], and it plays a structural role in our method: it allows σ^2 to be profiled out of the response-variable loss J_y , that arises as an optimisation subproblem in the minimisation step of the EM algorithm, resulting in a convex formulation, which guarantees a global rather than local minimum for J_y . This scaling, together with averaging $\ell_o(\theta)$ over the n observations, also reduces the dependence of the tuning parameters on both σ^2 and the sample size, making the process of choosing tuning parameters more stable across data sets.

Our method for estimating the parameters of the censored-covariate regression model involves minimising $\ell_o^{\text{pen}}(\theta)$; we describe how to do this minimisation in Section 2.2.

2.2 | Parameter Estimation with an EM Algorithm

Direct minimisation of $\ell_o^{\text{pen}}(\theta)$ is difficult because the contributions to $\ell_o(\theta)$ from observations with censored x_i involve integral expressions. To address this difficulty, we use an EM algorithm [29], which decouples the solving of the integral expressions from the minimisation over θ , resulting in two separate alternating steps. The EM algorithm for minimising $\ell_o^{\text{pen}}(\theta)$ uses the penalised complete-data loss function

$$\ell_c^{\text{pen}}(\theta) = -\frac{1}{n}\ell_c(\theta) + \frac{\lambda_1}{\sigma^2}\|\beta\|_1 + \lambda_2\|\gamma\|_1,$$

where it iteratively alternates between 1) the E-step: taking the expectation of $\ell_c^{\text{pen}}(\theta)$ conditioned on the observed data and the current parameter estimates, and 2) the M-step: minimising the resulting conditional expectation of $\ell_c^{\text{pen}}(\theta)$ to obtain new parameter estimates. Since the M-step as shown later is solved locally rather than globally, the resulting procedure is a *generalised* EM algorithm [30] and does not increase $\ell_o^{\text{pen}}(\theta)$.

Initialisation

We use the notation $\theta^{(k)}$ to denote the parameter estimates at the k^{th} iteration of the EM algorithm, where $k \geq 0$. When $k = 0$, $\theta^{(0)}$ represents the initial parameter estimates prior to the first E-step. We set the entries of $\theta^{(0)}$ as follows: $\alpha^{(0)} = 1$, $\beta^{(0)} = \mathbf{0}$, $\beta_0^{(0)} = 0$, $\sigma^{2(0)} = 1$, $\gamma^{(0)} = \mathbf{0}$, and $\gamma_0^{(0)} = 0$.

Expectation Step

Let $\mathcal{D}_i = [y_i \ x_i \ \mathbf{z}_i]$ for $\delta_i = 1$ and $\mathcal{D}_i = [y_i \ c_i \ \mathbf{z}_i]$ for $\delta_i = 0$ denote the observed data for index i , and let $\mathcal{D} = \{\mathcal{D}_i : i = 1, \dots, n\}$ denote the full observed data set. We define the Q -function to be the conditional expectation of the penalised complete-data loss function given $\mathcal{D}$ and the current parameter estimates $\theta^{(k)}$:

$$Q(\theta; \theta^{(k)}) := E\{\ell_c^{\text{pen}}(\theta) \mid \mathcal{D}; \theta^{(k)}\}.$$

Since the only unknown quantity is X_i , which only occurs in observations where $\delta_i = 0$, we use the linearity of the expectation

operator to write the Q -function as

$$\begin{aligned} Q(\theta; \theta^{(k)}) &= -\frac{1}{n} \sum_{i: \delta_i=1} \{\log f(y_i \mid x_i, \mathbf{z}_i; \theta) + \log f(x_i \mid \mathbf{z}_i; \theta)\} \\ &\quad - \frac{1}{n} \sum_{i: \delta_i=0} \left[E\{\log f(y_i \mid X_i, \mathbf{z}_i; \theta) \mid \mathcal{D}_i; \theta^{(k)}\} \right. \\ &\quad \left. + E\{\log f(X_i \mid \mathbf{z}_i; \theta) \mid \mathcal{D}_i; \theta^{(k)}\} \right] \\ &\quad + \frac{\lambda_1}{\sigma^2}\|\beta\|_1 + \lambda_2\|\gamma\|_1, \end{aligned}$$

where the independence of the observations implies that each conditional expectation depends only on the corresponding $\mathcal{D}_i$ rather than the full observed data set $\mathcal{D}$.

To evaluate the conditional expectations in $Q(\theta; \theta^{(k)})$, we need the density of the conditional random variable $X_i \mid \mathcal{D}_i; \theta^{(k)}$. We denote this density’s function by $t_i^{(k)}(x)$, and since $\mathcal{D}_i$ tells us that $X_i > c_i$ when $\delta_i = 0$, the density function is truncated to the interval (c_i, ∞) :

$$t_i^{(k)}(x) = \frac{f(y_i \mid x, \mathbf{z}_i; \theta^{(k)})f(x \mid \mathbf{z}_i; \theta^{(k)})}{\int_{c_i}^{\infty} f(y_i \mid x, \mathbf{z}_i; \theta^{(k)})f(x \mid \mathbf{z}_i; \theta^{(k)})dx}, \quad x > c_i.$$

This form of $t_i^{(k)}(x)$ is derived using the conditional non-informative censoring assumptions and an application of Bayes’ theorem. Plugging the normal and χ^2 density functions from (1) and (2) into $f(y_i \mid x, \mathbf{z}_i; \theta^{(k)})$ and $f(x \mid \mathbf{z}_i; \theta^{(k)})$, we have that $t_i^{(k)}(x)$ is proportional to

$$\exp\left\{-\frac{1}{2\sigma^{2(k)}}(y_i - \beta_0^{(k)} - \alpha^{(k)}x - \beta^{(k)\top}\mathbf{z}_i)^2 - \frac{x}{2}\right\} x^{\frac{1}{2}\exp(\gamma_0^{(k)} + \gamma^{(k)\top}\mathbf{z}_i) - 1},$$

which, when integrated, does not have a closed form since the χ^2 specification for X_i is not conjugate to the normal distribution for Y_i . Therefore, numerical integration techniques, such as the composite trapezoidal rule after determining an effective domain of integration, must be used to compute the normalising constant of $t_i^{(k)}(x)$.

When the conditional expectations in $Q(\theta; \theta^{(k)})$ are simplified (see Appendix A), the only expectation expressions that appear are

$$\begin{aligned} E(X_i^r \mid \mathcal{D}_i; \theta^{(k)}) &= \int_{c_i}^{\infty} x^r \cdot t_i^{(k)}(x) dx, \\ E(\log X_i \mid \mathcal{D}_i; \theta^{(k)}) &= \int_{c_i}^{\infty} \log x \cdot t_i^{(k)}(x) dx, \end{aligned}$$

where $r = 1, 2$. Similar to the normalising constant of $t_i^{(k)}(x)$, these expressions must be computed using numerical integration techniques. Once computed, we use these expressions to define the following quantities:

$$\begin{aligned} m_i^{(k)} &:= E(X_i \mid \mathcal{D}_i; \theta^{(k)}), \\ s_i^{(k)} &:= \sqrt{E(X_i^2 \mid \mathcal{D}_i; \theta^{(k)}) - E^2(X_i \mid \mathcal{D}_i; \theta^{(k)})}, \\ \ell_i^{(k)} &:= E(\log X_i \mid \mathcal{D}_i; \theta^{(k)}), \end{aligned}$$

which are required in the upcoming M-step.

Minimisation Step

We now show how to minimise the Q -function over θ . Apart from its computational role, the minimisation step is where the dual feature selection introduced in Section 2.1 becomes algorithmically tractable: as we show below, the Q -function separates into a loss function depending on $[\alpha, \beta, \beta_0, \sigma^2]$ and a loss function depending on $[\gamma, \gamma_0]$, which means the L_1 penalties on β and γ can be applied to their respective loss functions independently. Appendix A simplifies the Q -function and absorbs any terms that do not depend on θ into an additive constant $K^{(k)}$ so that it can be written as

$$Q(\theta; \theta^{(k)}) = J_y(\alpha, \beta, \beta_0, \sigma^2; \theta^{(k)}) + J_x(\gamma, \gamma_0; \theta^{(k)}) + K^{(k)},$$

which is separable in $[\alpha, \beta, \beta_0, \sigma^2]$ and $[\gamma, \gamma_0]$. Hence, the M-step reduces to minimising the two loss functions

$$\begin{aligned} J_y(\alpha, \beta, \beta_0, \sigma^2; \theta^{(k)}) &= \frac{1}{2n\sigma^2} (\|\mathbf{y} - \beta_0 \mathbf{1}_n - \alpha \mathbf{w}^{(k)} - \mathbf{Z}\beta\|_2^2 + \|\alpha \mathbf{s}^{(k)}\|_2^2) \\ &\quad + \frac{1}{2} \log \sigma^2 + \frac{\lambda_1}{\sigma^2} \|\beta\|_1, \\ J_x(\gamma, \gamma_0; \theta^{(k)}) &= \frac{1}{2n} \mathbf{b}^{(k)\top} \exp(\gamma_0 \mathbf{1}_n + \mathbf{Z}\gamma) \\ &\quad + \frac{1}{n} \mathbf{1}_n^\top \log \Gamma \left\{ \frac{1}{2} \exp(\gamma_0 \mathbf{1}_n + \mathbf{Z}\gamma) \right\} + \lambda_2 \|\gamma\|_1, \end{aligned}$$

where, after a notational reordering that places observations with complete x_i before observations with censored x_i , we define the stacked column vectors $\mathbf{y} = [\mathbf{y}_1, \mathbf{y}_0]$, $\mathbf{w}^{(k)} = [\mathbf{x}, \mathbf{m}^{(k)}]$, and $\mathbf{b}^{(k)} = [\log 2 - \log \mathbf{x}, \log 2 - \mathbf{e}^{(k)}]$, and the vertically-stacked matrix $\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_0]$. We also write $\mathbf{m}^{(k)} = [m_i^{(k)}]_{i:\delta_i=0}$, $\mathbf{s}^{(k)} = [s_i^{(k)}]_{i:\delta_i=0}$, $\mathbf{e}^{(k)} = [e_i^{(k)}]_{i:\delta_i=0}$, $\mathbf{x} = [x_i]_{i:\delta_i=1}$, as well as $\mathbf{y}_d = [y_i]_{i:\delta_i=d}$ and $\mathbf{Z}_d = [\mathbf{z}_i^\top]_{i:\delta_i=d}$ for $d = 0, 1$. The matrices $\mathbf{Z}_1$ and $\mathbf{Z}_0$ denote the feature matrices for observations with complete and censored x_i respectively. Scalar-input functions such as the exponential, logarithm, and Γ functions are applied to vectors entry-wise. The minimisers of J_y and J_x are collected together to form $\theta^{(k+1)}$, which is then used by the next E-step to compute $Q(\theta; \theta^{(k+1)})$.

To minimise J_y , we first solve

$$\arg \min_{\alpha, \beta, \beta_0} \left\{ \frac{1}{2n} (\|\mathbf{y} - \beta_0 \mathbf{1}_n - \alpha \mathbf{w}^{(k)} - \mathbf{Z}\beta\|_2^2 + \|\alpha \mathbf{s}^{(k)}\|_2^2) + \lambda_1 \|\beta\|_1 \right\}, \quad (4)$$

to obtain the solution $[\alpha^{(k+1)}, \beta^{(k+1)}, \beta_0^{(k+1)}]$, and then we use this solution to obtain

$$\begin{aligned} \sigma^{2(k+1)} &= \frac{1}{n} (\|\mathbf{y} - \beta_0^{(k+1)} \mathbf{1}_n - \alpha^{(k+1)} \mathbf{w}^{(k)} - \mathbf{Z}\beta^{(k+1)}\|_2^2 + \|\alpha^{(k+1)} \mathbf{s}^{(k)}\|_2^2) \\ &\quad + 2\lambda_1 \|\beta^{(k+1)}\|_1. \end{aligned}$$

These $k+1^{\text{th}}$ -iteration parameter estimates together minimise J_y globally, not merely locally, as shown by a profiled-loss-function argument in Appendix B. The global minimisation is the payoff of scaling λ_1 by $1/\sigma^2$ in $\ell_o^{\text{pen}}(\theta)$, which allows σ^2 to be profiled out and leaves the penalised least-squares problem shown in (4). This penalised least-squares problem can be seen as a generalised elastic net [31] problem because we have an L_1 penalty on β with strength λ_1 , and an L_2 penalty on α with strength $\|\mathbf{s}^{(k)}\|_2^2$. To solve this optimisation problem, we may use the `glmnet` R package

from Sokolov [32], which implements a cyclic coordinate descent algorithm.

Unlike J_y , J_x is non-convex, so we can only guarantee a local minimum. To minimise J_x , we first write $J_x(\gamma, \gamma_0; \theta^{(k)}) = g_x(\gamma_0, \gamma; \theta^{(k)}) + \lambda_2 \|\gamma\|_1$, where g_x denotes the smooth part. Then, we exploit this smooth-plus- L_1 structure to attain a minimum by using the orthant-wise limited-memory quasi-Newton (OWL-QN) algorithm implemented in the `lbfgs` R package from Copola et al. [33], initialised at $[\gamma_0^{(k)}, \gamma^{(k)}]$. OWL-QN requires the gradient of the smooth part, $\nabla g_x(\gamma_0, \gamma; \theta^{(k)})$, which is given by

$$\frac{1}{2n} [\mathbf{1}_n \mathbf{Z}]^\top \left(\exp(\gamma_0 \mathbf{1}_n + \mathbf{Z}\gamma) \odot \left[\mathbf{b}^{(k)} + \psi \left\{ \frac{1}{2} \exp(\gamma_0 \mathbf{1}_n + \mathbf{Z}\gamma) \right\} \right] \right),$$

where $\odot$ denotes entry-wise multiplication, ψ is the digamma function, and $[\mathbf{1}_n \mathbf{Z}]$ is the feature matrix with a vector of ones as the first column. The resulting minimiser is taken to be $[\gamma_0^{(k+1)}, \gamma^{(k+1)}]$.

Computational Considerations

Two computational considerations are worth noting. First, we declare convergence when the percentage change in the penalised observed-data loss function $\ell_o^{\text{pen}}(\theta)$ falls below a small threshold τ , i.e. if $\Delta^{(k+1)} := 100 \left| \left\{ \ell_o^{\text{pen}}(\theta^{(k+1)}) - \ell_o^{\text{pen}}(\theta^{(k)}) \right\} / \ell_o^{\text{pen}}(\theta^{(k)}) \right|$, then the EM algorithm stops when $\Delta^{(k+1)} < \tau$. Second, before parameter estimation, we standardise the response variable y and each column of $\mathbf{Z}$ to help with numerical stability. Standardisation of the columns of $\mathbf{Z}$ also makes the L_1 penalties act fairly across the entries of β and γ . Appendix C provides the exact standardisation scheme together with the back-transformation used to report parameter estimates on the original-data scale.

2.3 | Tuning Parameter Selection

Our method uses two tuning parameters, λ_1 and λ_2 , which control the amount of sparsity in the β and γ coefficient vectors respectively. Choosing a λ_1 and λ_2 in a data-driven manner is not straightforward as there are several possible approaches, including information-criteria-based approaches and cross-validation-based approaches. Information criteria such as the AIC [34], BIC [35], and other criteria [36] are generally rooted in likelihood theory and require specific assumptions about the correlation between features and how the sample size grows, which we do not make. Furthermore, since we use L_1 penalties that do shrinkage as well as selection, it is not clear how to compute the degrees of freedom to use in an information criterion. K -fold cross-validation (CV), whilst more computationally expensive as it requires fitting models on multiple subsets of the data, is a model-agnostic approach that does not require specific assumptions and simply focuses on optimising a prediction error metric. Therefore, we adopt a CV approach for choosing λ_1 and λ_2 .

Since we hold scientific interest in both the response variable y and the censored covariate x , the prediction error metric to use in CV must account for both. To formulate such a metric, one could use some sort of weighted average that combines the prediction error of both y and x together. However, it is not clear how the weights should be set in this sort of metric. Instead, we decide to

use a mean absolute error (MAE) metric based on prediction error of y where we first predict x and then plug that prediction into the response-variable model component to obtain a prediction of y . This metric prioritises prediction of the response variable. It therefore does not necessarily optimise prediction or feature recovery for the censored-covariate component considered in isolation. We adopt it because the response is the primary prediction target in the regression model, while allowing prediction error in the censored-covariate component to influence tuning through the plug-in prediction. Specifically, we define the cross-validation mean absolute error (CV MAE),

$$\text{CV MAE}(\lambda_1, \lambda_2) = \frac{1}{K} \sum_{k=1}^K \left\{ \frac{1}{n_k} \sum_{i \in \mathcal{F}_k} |y_i - \hat{y}_i^{(-k)}(\lambda_1, \lambda_2)| \right\} \quad (5)$$

where $\mathcal{F}_k$ denotes the set of indices in fold k , $n_k = |\mathcal{F}_k|$, and $\hat{y}_i^{(-k)}(\lambda_1, \lambda_2)$ is the prediction from the response-variable model component of the censored-covariate regression model fitted on all observations outside fold k . To form this prediction for a given λ_1 and λ_2 , we compute

$$\begin{aligned} \hat{x}_i^{(-k)} &= \exp(\hat{\gamma}_0^{(-k)} + \hat{\gamma}^{(-k)\top} \mathbf{z}_i), \\ \hat{y}_i^{(-k)} &= \hat{\beta}_0^{(-k)} + \hat{\alpha}^{(-k)} \hat{x}_i^{(-k)} + \hat{\beta}^{(-k)\top} \mathbf{z}_i, \end{aligned}$$

where, to avoid clutter, each hatted symbol's dependence on λ_1 and λ_2 is not explicitly written. In computing $\hat{y}_i^{(-k)}$, we use $\hat{x}_i^{(-k)}$ for every index in the held-out fold, regardless of whether x_i is complete or censored, so that prediction error of x_i propagates into the prediction error of y_i . By default, we use $K = 5$ folds.

When randomly distributing the indices across the folds, we stratify by the censoring indicator δ_i , so that each fold contains approximately the same proportion of observations with censored x_i as the full data set. This stratification prevents some folds from containing too few observations with censored x_i , which would otherwise make the CV MAE for those folds unstable and less representative of out-of-sample performance.

In choosing λ_1 and λ_2 , we compute the CV MAE over a two-dimensional grid of candidate λ_1 and λ_2 values, which we take to be of size 30×30 . Two design choices from Section 2.1 make it easier to set the endpoints of this grid. Since λ_1 and λ_2 do not depend on n and σ^2 , and since the response variable and features are standardised before model fitting, the choice $[\lambda_1, \lambda_2] = [1, 1]$ gives $[\hat{\beta}, \hat{\gamma}] \approx \mathbf{0}$, which we take to be the top-right endpoint of the grid. The bottom-left endpoint is chosen to be $[0.001, 0.001]$; we find that such values of λ_1 and λ_2 typically correspond to all features being contained in the sets selected for the response-variable model component and the censored-covariate model component. Between these endpoints, we space the grid logarithmically, which allows us to have finer resolution in the low-penalty region where the selected sets of features tend to change most rapidly.

Although taking λ_1 and λ_2 that minimise the CV MAE is a natural choice, this choice tries to optimise for out-of-sample prediction error, so it may not select the most parsimonious sets of features. Therefore, we also consider a two-tuning-parameter analogue of the one standard error (1-SE) rule used in the `glmnet` R package [37], which is a heuristic for selecting a smaller set of features in the lasso. Let $[\hat{\lambda}_1^{\min}, \hat{\lambda}_2^{\min}]$ denote the values of λ_1 and λ_2 that

minimise the CV MAE over the grid, and let $\text{SD}(\hat{\lambda}_1^{\min}, \hat{\lambda}_2^{\min})/\sqrt{K}$ be the standard error of the CV MAE across the K folds. Then, out of all λ_1 and λ_2 that satisfy

$$\text{CV MAE}(\lambda_1, \lambda_2) \leq \text{CV MAE}(\hat{\lambda}_1^{\min}, \hat{\lambda}_2^{\min}) + \text{SD}(\hat{\lambda}_1^{\min}, \hat{\lambda}_2^{\min})/\sqrt{K},$$

choose λ_1 and λ_2 that correspond to the censored-covariate regression model with the fewest number of non-zero entries across $\hat{\beta}$ and $\hat{\gamma}$ when fitted on the full data set. This analogue mirrors the usual 1-SE principle by preferring a simpler model whose out-of-sample prediction error is statistically indistinguishable from that of the model where λ_1 and λ_2 are selected to minimise the CV MAE.

2.4 | Software Implementation

The proposed method is implemented in the R package `ccreds` (Censored Covariate Regression with Dual Selection), available at <https://github.com/rajan-shankar/ccreds/>. The package allows users to fit censored-covariate regression models over a grid of $[\lambda_1, \lambda_2]$ pairs, evaluating CV MAE scores across K folds with a helper function to extract the best model according to either the minimum CV MAE rule or the 1-SE rule. The package also provides functions for producing visual diagnostics such as CV MAE heat maps and coefficient path plots.

3 | Simulation Study

We claim that, in finite-sample settings with correlated features and adequate signal in the data, our method can recover the active sets of features for both the response-variable model component and the censored-covariate model component, even when those active sets differ. By ‘adequate signal’, we mean an adequate signal-to-noise ratio. The purpose of this simulation study is to assess this claim by intentionally having different active sets of features be associated with the two model components. It is not intended to assess weak-signal settings; the latter is considered separately in Section 4.2.

3.1 | Simulation Set-up

Denote the *active* sets of features for the response-variable model component and the censored-covariate model component as

$$\begin{aligned} \mathcal{A}_y &:= \{z_j : j = 1, \dots, p, \beta_j \neq 0\} \quad \text{and} \\ \mathcal{A}_x &:= \{z_j : j = 1, \dots, p, \gamma_j \neq 0\} \end{aligned}$$

respectively, where β_j and γ_j with $j = 1, \dots, p$ are entries of the ground-truth coefficient vectors β and γ . We simulate data sets with $n = 1000$ observations and $p = 30$ features, and we consider four different simulation settings by fixing $\mathcal{A}_y = \{z_j, z_{31-j} : j = 1, 2, 3, 4\}$ and varying $\mathcal{A}_x$:

- *Exactly same:* $\mathcal{A}_x = \{z_j, z_{31-j} : j = 1, 2, 3, 4\}$. The two active sets are the same.
- *Mutually exclusive:* $\mathcal{A}_x = \{z_j, z_{31-j} : j = 5, 6, 7, 8\}$. The two active sets do not have any features in common.

- *Some similarity:* $\mathcal{A}_x = \{z_j, z_{31-j} : j = 3, 4, 5, 6\}$. There is partial overlap between the two active sets.
- *All $\gamma = 0$:* $\mathcal{A}_x = \emptyset$. No features are associated with x , so the censored-covariate model component is simply an intercept-only model.

We generate $B = 100$ simulated data sets under each of the four simulation settings.

For each observation in a simulated data set, we generate features from a $N_p(\mathbf{0}, \Sigma)$ distribution, where Σ is a Toeplitz covariance matrix with an autoregressive dependence structure between its entries, given by $\Sigma_{jk} = 9 \times \rho^{|j-k|}$ with $\rho = 0.5$ and $j, k = 1, \dots, p$. We set the β -coefficients of the first four features in $\mathcal{A}_y$ to 4, 4, 2, 2 and those of the last four features to -4, -4, -2, -2. Then, excluding the all $\gamma = 0$ setting, we set the γ -coefficients of the first four features in $\mathcal{A}_x$ to 0.04, 0.04, 0.02, 0.02 and those of the last four features to -0.04, -0.04, -0.02, -0.02. The γ -coefficients are much smaller than the β -coefficients to balance out the exponential function in the mean of the χ^2 distribution. For the other parameters of the censored-covariate regression model, we set $\alpha = 10$, $\beta_0 = 5$, $\gamma_0 = 2$, and $\sigma^2 = 100$.

Plugging the parameters and feature values into (1) and (2), we generate values of y_i and x_i for $i = 1, \dots, n$. After generating x_i , we generate c_i so that $\frac{1}{n} \sum_{i=1}^n 1_{\{c_i < x_i\}} \approx r_{\text{cens}}$, where r_{cens} is the desired censoring rate. Specifically, if we generate x_i from $X_i | \mathbf{Z}_i = \mathbf{z}_i \sim \chi^2(\mu_i)$ with $\mu_i = \exp(\gamma_0 + \boldsymbol{\gamma}^\top \mathbf{z}_i)$, then we generate c_i from $C_i | \mathbf{Z}_i = \mathbf{z}_i \sim \chi^2(\nu_i)$ with ν_i chosen so that $P(C_i < X_i | \mathbf{Z}_i = \mathbf{z}_i) = r_{\text{cens}}$. The details of how ν_i is chosen are left to Appendix D. For this simulation study, we set $r_{\text{cens}} = 50\%$. Whenever we have $c_i < x_i$, only c_i is retained and the censoring indicator δ_i is set to 0 for those observations.

We evaluate four models fitted using our method, chosen so that comparisons between them isolate different aspects of performance. Two of these models are the *min-CV* and *1-SE* models, which we define to be the models fitted when using λ_1 and λ_2 that minimise the CV MAE, and when using λ_1 and λ_2 chosen by the 1-SE rule, respectively; more details on these ways to choose λ_1 and λ_2 are available in Section 2.3. Comparing them tests whether the more parsimonious 1-SE rule meaningfully changes selection performance relative to the prediction-error-minimising min-CV rule. We also consider the *1-SE post-selection* model, which fits the censored-covariate regression model using the sets of features selected by the 1-SE model. Compared to the 1-SE model, this model reduces shrinkage bias in the estimated coefficient vectors after the sets of features have been selected. Finally, we consider the *oracle* model, which is given the active sets of features used to generate the simulated data set, and fits the censored-covariate regression model so that the response-variable model component only uses features in $\mathcal{A}_y$ and the censored-covariate model component only uses features in $\mathcal{A}_x$. The oracle model is not available in practice, but it provides a useful benchmark for how well the other models perform relative to a best-case scenario where the active sets are known in advance.

3.2 | Evaluation Metrics

To evaluate the models across the simulation settings, we compute two types of metrics—selection performance and estimation error—separately for the response-variable model component and the censored-covariate model component. Separating the metrics by model component lets us assess whether our method is doing well on each individual component, rather than averaging metrics across the two components into a single number that hides imbalances.

Selection Performance Metrics

For a given model, let $\hat{\boldsymbol{\beta}}^{(b)}$ and $\hat{\boldsymbol{\gamma}}^{(b)}$ denote the estimated coefficient vectors from simulated data set b , where $b = 1, \dots, B$ and $B = 100$. Two metrics that we use for evaluating selection performance are the sensitivity and specificity. For the response-variable model component, we compute

$$\text{Sens}_y^{(b)} = \frac{\sum_{j=1}^p 1_{\{\hat{\beta}_j^{(b)} \neq 0, \beta_j \neq 0\}}}{\sum_{j=1}^p 1_{\{\beta_j \neq 0\}}} \quad \text{and} \quad \text{Spec}_y^{(b)} = \frac{\sum_{j=1}^p 1_{\{\hat{\beta}_j^{(b)} = 0, \beta_j = 0\}}}{\sum_{j=1}^p 1_{\{\beta_j = 0\}}},$$

with analogous definitions $\text{Sens}_x^{(b)}$ and $\text{Spec}_x^{(b)}$ for the censored-covariate model component obtained by replacing β_j and $\hat{\beta}_j^{(b)}$ with γ_j and $\hat{\gamma}_j^{(b)}$. We report these metrics after averaging over the B simulated data sets. Note that, in the all $\gamma = 0$ setting, sensitivity for the censored-covariate model component is not defined because $\sum_{j=1}^p 1_{\{\gamma_j \neq 0\}} = 0$, so we report only the specificity for this model component in this simulation setting.

We also consider a feature-specific selection performance metric, where for each feature z_j and for a specific model component, we define its selection frequency to be the average proportion of simulated data sets for which that feature is contained in the selected set for that model component, namely

$$\text{SF}_y(z_j) = \frac{1}{B} \sum_{b=1}^B 1_{\{\hat{\beta}_j^{(b)} \neq 0\}} \quad \text{and} \quad \text{SF}_x(z_j) = \frac{1}{B} \sum_{b=1}^B 1_{\{\hat{\gamma}_j^{(b)} \neq 0\}}$$

for the response-variable model component and the censored-covariate model component respectively.

Estimation Error Metrics

To complement the selection performance metrics, we quantify how close the estimated coefficient vectors are to the ground-truth coefficient vectors for each model component. For simulated data set b , we compute the relative L_2 estimation errors

$$\text{RelL}_{2,\beta}^{(b)} = \frac{\|\hat{\boldsymbol{\beta}}^{(b)} - \boldsymbol{\beta}\|_2}{\|\boldsymbol{\beta}\|_2} \quad \text{and} \quad \text{RelL}_{2,\gamma}^{(b)} = \frac{\|\hat{\boldsymbol{\gamma}}^{(b)} - \boldsymbol{\gamma}\|_2}{\|\boldsymbol{\gamma}\|_2},$$

which we then average over the B simulated data sets. In the all $\gamma = 0$ setting, $\text{RelL}_{2,\gamma}^{(b)}$ is not defined because $\|\boldsymbol{\gamma}\|_2 = 0$, so we compute only $\text{RelL}_{2,\beta}^{(b)}$ for this simulation setting.

TABLE 1 | Sensitivity and specificity (%) for the response-variable model component (y) and censored-covariate model component (x) across the four simulation settings, averaged over $B = 100$ simulation data sets. Results are shown for the min-CV and 1-SE models.

Model component	Setting	Min-CV		1-SE	
		Sens.	Spec.	Sens.	Spec.
Response variable	Exactly same	100.0	39.3	99.9	99.5
	Mutually exclusive	100.0	55.5	100.0	99.9
	Some similarity	100.0	46.5	100.0	99.6
	All $\gamma = 0$	100.0	56.2	100.0	100.0
Censored covariate	Exactly same	100.0	71.5	99.1	98.7
	Mutually exclusive	99.9	71.9	97.4	98.2
	Some similarity	99.8	68.2	96.8	97.7
	All $\gamma = 0$	—	97.0	—	100.0

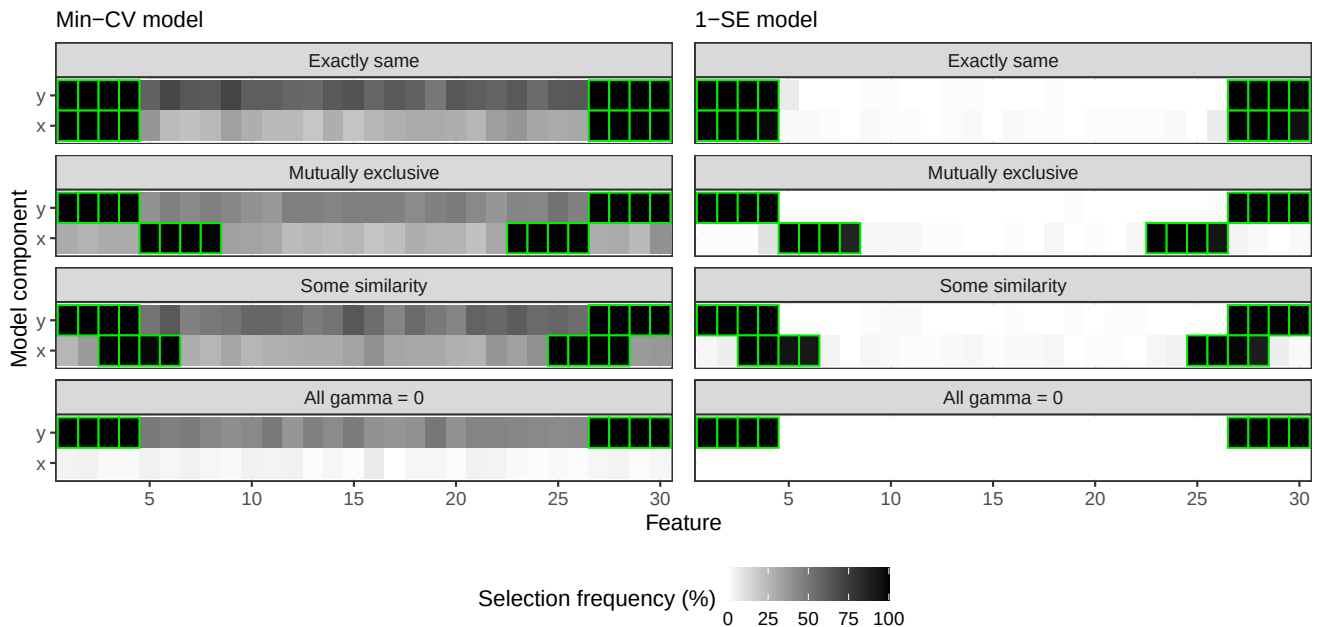


FIGURE 1 | Selection-frequency heat maps across all features $z_1, \dots, z_{30}$ in the four simulation settings. The active sets $\mathcal{A}_y$ and $\mathcal{A}_x$ in each simulation setting are outlined in green. Results are shown for the min-CV and 1-SE models.

3.3 | Results

Table 1 and Figure 1 show that our method is effective at recovering the active sets of features for both the response-variable model component and the censored-covariate model component. Across all four simulation settings, both the min-CV and 1-SE models achieve very high sensitivity on both model components, and the selection-frequency heat maps show that the features in the active sets are contained in the selected sets for almost all of the simulated data sets. Between the two tuning rules, the 1-SE model achieves higher specificity and thus correctly excludes features not in the active sets from the selected sets most of the time, whereas the min-CV model's smaller choice of λ_1 and λ_2 cause it to incorrectly include more of these features in the selected sets for many of the simulated data sets. We note that Table 1 and Figure 1 report selection performance results only for the min-CV and 1-SE models, since the oracle model has perfect sensitivity and specificity by construction and the 1-SE post-selection model uses the same sets of features as those selected by the 1-SE model.

Despite the 1-SE model's better selection performance, Table 2 shows that it has slightly higher relative L_2 estimation error than the min-CV model, on both model components. This result is consistent with the larger λ_1 and λ_2 chosen by the 1-SE rule, which introduces greater shrinkage bias into the estimated coefficient vectors. The 1-SE post-selection model reduces this bias by fitting the censored-covariate regression model on the sets of features selected by the 1-SE model, and consequently its relative L_2 estimation error is lower than that of the min-CV model and close to the oracle model.

Across the four simulation settings, there is little variation in each of the evaluation metrics, indicating that our method is relatively insensitive to how the features are distributed across $\mathcal{A}_y$ and $\mathcal{A}_x$. The only exception is the all $\gamma = 0$ setting, where in Table 1, the specificity for the censored-covariate model component is higher than in the other simulation settings. This result is expected because, in the other simulation settings where $\mathcal{A}_x \neq \emptyset$, the features that are not in $\mathcal{A}_x$ but are correlated with those in $\mathcal{A}_x$ are more likely to be in the selected set for the censored-covariate

TABLE 2 | Relative L_2 estimation errors for the response-variable model component (y) and censored-covariate model component (x) across the four simulation settings, averaged over $B = 100$ simulated data sets. Results are shown for the min-CV, 1-SE, 1-SE post-selection, and oracle models.

Coef.	Setting	Min-CV	1-SE	1-SE post-sel.	Oracle
β	Exactly same	0.08	0.11	0.06	0.05
	Mutually excl.	0.07	0.09	0.05	0.05
	Some similarity	0.08	0.09	0.05	0.05
	All $\gamma = 0$	0.08	0.09	0.05	0.05
γ	Exactly same	0.24	0.26	0.21	0.19
	Mutually excl.	0.25	0.29	0.23	0.19
	Some similarity	0.25	0.29	0.23	0.19
	All $\gamma = 0$	—	—	—	—

model component, resulting in lower specificity for that model component.

In comparing the evaluation metrics between the response-variable model component and the censored-covariate model component, Table 2 shows higher relative L_2 estimation error for γ than for β across all models and simulation settings. This result is likely due to a combination of weaker signal for the censored-covariate model component, and the γ -coefficients of features in $\mathcal{A}_x$ being much smaller than the β -coefficients of features in $\mathcal{A}_y$, making γ harder to estimate.

In simulations not presented here, the selection performance metrics remain similar across increasing censoring rates from 25% to 75%, while the relative L_2 estimation error increases slightly.

4 | Application to Huntington Disease Data

In this section, we first construct the Huntington disease data set and use our method to do feature selection, reporting the results. Second, we run a simulation study on artificial data sets constructed to mimic the real HD data set, allowing us to assess the stability of the selected sets of features from the real-data application.

4.1 | Application

We use our method to select sets of brain regions that are associated with Symbol Digit Modalities Test (SDMT) score and time-to-diagnosis (TTD) of HD on the PREDICT-HD data introduced in Section 1.2. We curate a data set of $n = 692$ individuals, retaining only their measurements taken at study entry. We compute TTD as follows. For each individual i , let $A_{\text{Entry},i}$ be their age at study entry, $A_{\text{Exit},i}$ be their age when they exited the study, and $A_{\text{Diag},i}$ be their age at the visit when they received a Diagnostic Confidence Level (DCL) of 4, i.e. were diagnosed with HD. If individual i was never diagnosed during the study, we set $A_{\text{Diag},i} = \infty$. We define TTD as

$$\text{TTD}_i = \min(\{A_{\text{Diag},i}, A_{\text{Exit},i}\}) - A_{\text{Entry},i},$$

where, for individuals who were never diagnosed during the study, their TTD is right-censored. In this data set, 77.5% of individuals have a censored TTD, with the average censored TTD

for these individuals being 3.5 years. By using TTD, we are reverse-aligning [17] individuals by their age at diagnosis. Reverse alignment, compared to the usual alignment of individuals in survival studies by their age at entry, places individuals on a common disease-progression time scale, which makes prognostic patterns such as changes in brain-region volumes and cognitive ability easier to detect.

We fit a censored-covariate regression model with SDMT score as the response variable, TTD as the censored covariate, and $p = 40$ subcortical brain-region volumes obtained from a volumetric MRI scan as the features. A small number of these volumes are non-regional volumes, such as cerebrospinal fluid and white matter hypointensities, but for brevity we will refer to each of the 40 features as brain-region volumes. The brain-region volumes are normalised by head size by expressing them as percentages of the total volume inside the skull, which helps adjust for the fact that bigger heads have bigger brain-region volumes. We include three demographic variables—sex, education, and CAP (CAG-Age Product) score—in both model components as unpenalised features, meaning that their coefficients are estimated but are not allowed to be shrunk to 0. These demographic variables are included because they are clinically known to be associated with SDMT score and TTD [38, 39, 40]. For our analysis, this means that brain-region selection is conditional on these known variables: a brain region is selected only if it contributes additional information for SDMT score or TTD beyond sex, education, and CAP score, rather than acting as a proxy for them.

Figure 2 shows how the entries in the estimated coefficient vectors shrink to 0 as λ_1 and λ_2 increase. We examined both tuning rules. The 1-SE rule, which prioritises sparsity and specificity, selected no brain regions in either model component. The min-CV rule, which prioritises predictive performance and is less conservative under weak signal, selected non-empty candidate sets. We therefore report the min-CV results as an exploratory analysis, while treating the empty 1-SE result as evidence that the region-specific signals are not strong under a conservative selection rule. The stability analysis in Section 4.2 shows that models selected by the min-CV rule have much higher sensitivity than the those selected by the 1-SE rule in artificial data sets with similarly weak signal.

Since there is generally high (> 0.8) correlation between the left and right versions of a brain region, we are not too concerned about which of these versions is included in the selected sets; as such, the following interpretations of our results are based on whether any version of a brain region is included.

Table 3 reports the selected sets of brain regions that are associated with SDMT score and TTD, based on the min-CV model. The main result is that the two selected sets are partly distinct: 13 brain regions are selected for TTD, 5 are selected for SDMT score, and only cerebrospinal fluid and white matter hypointensities are selected for both model components. This result suggests that most of the selected brain-region information is associated with TTD, while a smaller subset has an additional direct association with SDMT score after conditioning on TTD. The selected set for SDMT score must be interpreted as having an additional effect after conditioning on TTD since the response-variable model component in (1) is a function of the censored covariate.

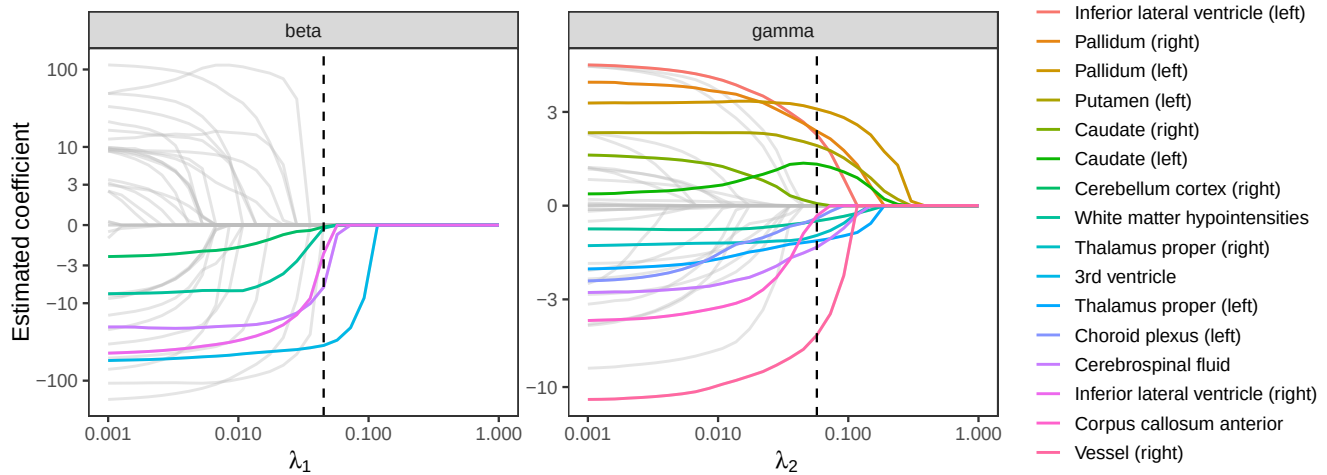


FIGURE 2 | Estimated β - and γ -coefficient paths for each brain region in the Huntington disease application as λ_1 and λ_2 vary over the log-spaced interval $[0.001, 1]$. The tuning-parameter values chosen by the min-CV model are $\lambda_1 = 0.045$ and $\lambda_2 = 0.057$, as indicated by the dashed vertical lines. The left plot varies λ_1 while keeping λ_2 fixed at its chosen value, and the right plot varies λ_2 while keeping λ_1 fixed at its chosen value. The greyed-out paths represent brain regions that were not in the selected set for the respective model component of the min-CV model.

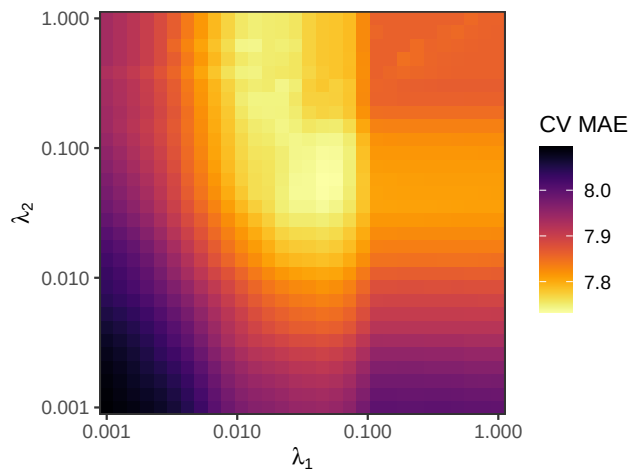


FIGURE 3 | Cross-validation mean absolute error (CV MAE) over the log-spaced tuning-parameter grid in the Huntington disease application. The min-CV model chooses $\lambda_1 = 0.045$ and $\lambda_2 = 0.057$.

TABLE 3 | Brain regions included in the selected set for SDMT score and/or time-to-diagnosis by the min-CV model in the Huntington disease application. A non-zero β -coefficient means that the brain region was in the selected set for SDMT score, while a non-zero γ -coefficient means that the brain region was in the selected set for time-to-diagnosis.

Brain region	β -coefficient	γ -coefficient
Caudate (left)	—	1.11
Caudate (right)	—	0.05
Choroid plexus (left)	—	-0.33
Cerebellum cortex (right)	-0.11	—
Inferior lateral ventricle (left)	—	2.07
Inferior lateral ventricle (right)	-1.88	—
Pallidum (left)	—	3.14
Pallidum (right)	—	2.20
Putamen (left)	—	1.69
Thalamus proper (left)	—	-0.93
Thalamus proper (right)	—	-0.78
Vessel (right)	—	-5.02
Corpus callosum anterior	—	-0.26
White matter hypointensities	-0.19	-0.39
Cerebrospinal fluid	-6.13	-1.11
3rd ventricle	-35.24	—

Focusing on brain regions that are exclusively in the selected set for TTD, we see that the caudate, putamen, and pallidum—brain regions that form the *basal ganglia*—are positively associated with TTD, which aligns with clinical research [41, 42]. The thalamus, which is associated with the basal ganglia, instead is negatively associated with TTD. We claim that a larger thalamus could be a sign that other basal ganglia are proportionally smaller, therefore being associated with less TTD.

We see that the cerebellum cortex is exclusively associated with SDMT score. It is known that the cerebellum, which includes the cerebellum cortex, is positively linked to cognitive ability [43]; however, Table 3 reports a negative β -coefficient for this brain region. We do not claim that the cerebellum cortex volume negatively correlates with cognitive ability; rather, this discrepancy may reflect adjustment for other, correlated brain regions

variables in the fitted model. We also see that the 3rd ventricle and the inferior lateral ventricle are exclusively associated with SDMT score. Ventricles could be associated with decline in cognitive ability; for example, Wittens et al. [44] show that the inferior-lateral-ventricle-to-hippocampus volume ratio correlates with this decline.

There are only two brain regions—cerebrospinal fluid and white matter hypointensities—that are in the selected set for both SDMT score and TTD. A large cerebrospinal-fluid volume indicates that other brain regions are proportionally smaller [45, 46], and white matter hypointensities are lesions that indicate brain damage or disease [47]. Both of these volumes are markers of general brain atrophy, and we claim that they are associated with both time until diagnosis and cognitive ability.

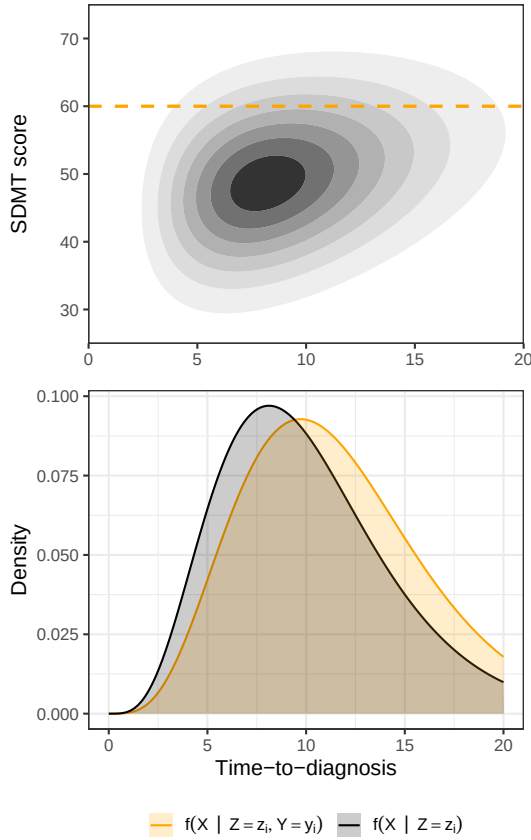


FIGURE 4 | *Top*: Bivariate density of SDMT score and time-to-diagnosis (TTD) for a representative out-of-sample individual from the fitted censored-covariate regression model. *Bottom*: Marginal and conditional densities of TTD for the out-of-sample individual. Both TTD densities condition on brain regions and demographic variables, but the conditional density additionally conditions on the individual's SDMT score.

Overall, these results show that our method can separate brain regions whose information is mainly about TTD from those with an additional association with SDMT score, producing clinically-plausible hypotheses for follow-up studies rather than a definitive neurobiological map. Since the brain-region volumes are correlated, the signs and magnitudes of individual coefficients should be interpreted as conditional, model-adjusted effects rather than as isolated, region-specific effects. We remark that our results are based on the min-CV model, although a censored-covariate regression model fitted using different values of λ_1 and λ_2 could be more clinically appropriate. For example, a clinician could combine their expertise with the estimated coefficient paths in Figure 2 and the CV MAE heat map in Figure 3 to arrive at a clinically-informed choice of λ_1 and λ_2 .

Individual-Level Prediction

In addition to feature selection, the fitted censored-covariate regression model also provides individual-specific distributional predictions for SDMT score and TTD. By taking an individual's brain-region data from a volumetric MRI scan, as well as their sex, education, and CAP score, and plugging these values as $\mathbf{z}_i$

into $f(y | x, \mathbf{z}_i; \theta)f(x | \mathbf{z}_i; \theta)$, we can plot the bivariate density of their predicted SDMT score and TTD, as shown in the top plot of Figure 4. The high-density area of this plot summarises the model-implied association between SDMT score and TTD for an individual with the specified baseline characteristics. We can also plot $f(x | \mathbf{z}_i; \theta)$, the marginal density of TTD for an individual, to understand how long they likely have until they are diagnosed with HD, as shown by the black density in the bottom plot of Figure 4. Furthermore, if we can obtain not only an individual's $\mathbf{z}_i$, but also their SDMT score y_i , then we can plot the conditional density of TTD, $f(x | y_i, \mathbf{z}_i; \theta)$, by normalising the bivariate density evaluated at y_i . This conditional density is shown as the orange density in the bottom plot of Figure 4. The conditional density incorporates the individual's current cognitive ability via their SDMT score and may therefore provide a more individualised prediction than the marginal density. For example, the individual represented in Figure 4 had a higher SDMT score than expected given their baseline characteristics. Under the fitted model, the higher observed SDMT score shifts the conditional density of TTD towards larger values, indicating a slightly longer expected time until diagnosis of the disease.

4.2 | Stability Analysis on Artificial Data

In this subsection, we analyse the stability of the results obtained in Section 4.1. Although the results showed that our method was able to select clinically relevant sets of features in the HD data set, we want to assess how far these sets of features should be trusted because our method may select slightly different sets of features on different perturbations of the data set, if there is not strong signal in the data set. Therefore, we conduct a simulation study where we simulate $B = 100$ artificial data sets that mimic the real HD data set, and evaluate the selection performance of our method.

Each artificial data set contains $n = 692$ observations (representing individuals) and $p = 40$ features (representing brain regions). Three demographic variables representing sex, education, and CAP score, are included as well. For each observation in an artificial data set, we generate the features and the demographic variables from a $N_{p+3}(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$ distribution, where $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ are the estimated mean vector and estimated covariance matrix of the combined vector of features and demographic variables from the real HD data set. We set the ground-truth β - and γ -coefficients, as well as the other parameters α , β_0 , σ^2 , and γ_0 , to be those of the min-CV post-selection model fitted on the real HD data set. The min-CV post-selection model is the model re-fit to the real HD data set using the sets of features selected by the min-CV model; hence, the active sets of features in this simulation study are the sets selected by the min-CV model. By using the min-CV post-selection model's estimated coefficients instead of those of the min-CV model, we avoid shrinkage bias, allowing us to maintain a similar amount of signal in the artificial data sets as in the real HD data set. We then plug this model's parameters into (1) and (2) to generate values of y_i (representing SDMT score) and x_i (representing TTD), where we control the probability that x_i is right-censored to be the same as the censoring rate in the real HD data set, i.e. $r_{\text{cens}} = 77.5\%$.

We evaluate both the min-CV and 1-SE models using the selection performance metrics of Section 3.2. Figure 5 shows that, in

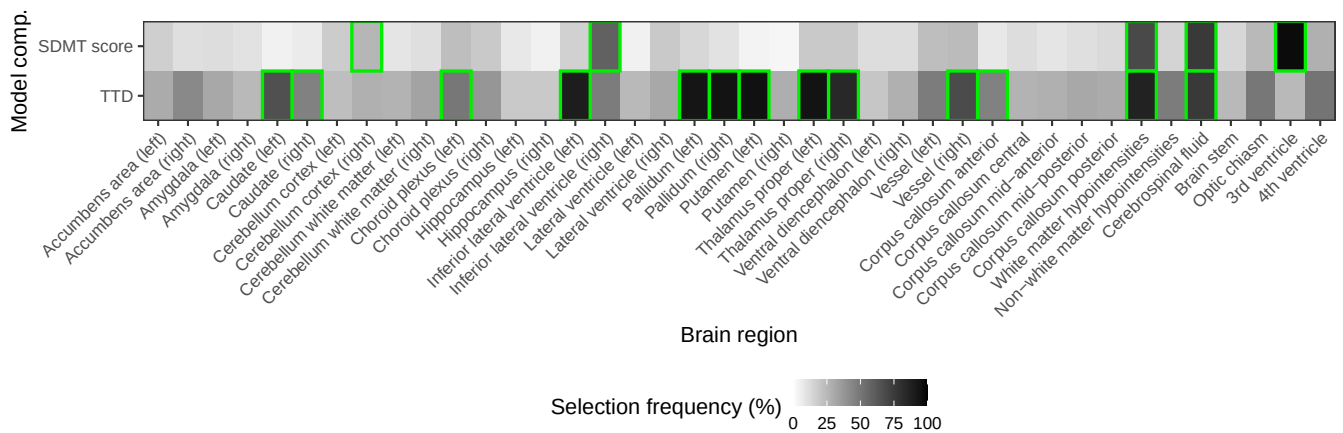


FIGURE 5 | Selection-frequency heat map for the artificial-data simulation study across the $p = 40$ brain regions, for the min-CV model. The active sets induced by the min-CV post-selection model are outlined in green.

TABLE 4 | Sensitivity and specificity (%) for the SDMT-score model component and time-to-diagnosis (TTD) model component in the artificial-data simulation study, averaged over $B = 100$ artificial data sets. Results are shown for the min-CV and 1-SE models.

Model component	Min-CV		1-SE	
	Sens.	Spec.	Sens.	Spec.
SDMT score (y)	65.4	86.6	3.2	99.5
Time-to-diagnosis (x)	76.0	68.1	4.2	99.4

the min-CV model, the selection frequencies of active-set features are mostly high, but not much higher than those of other features. The 1-SE model (not shown) is overly parsimonious and has $\leq 7\%$ selection frequency for all brain regions in both model components. Table 4 shows that the sensitivity across both model components, is not as high as in Table 1 for the min-CV model, and is near 0 for the 1-SE model, suggesting that the selected sets are not very stable. Since we saw in Section 3 that our method had good selection performance in data sets with adequate signal, this instability likely arises from the HD data set having weak signal, in that many brain regions do not correlate strongly with SDMT score and TTD, and from high correlation between brain regions. Many pairs of brain regions have correlation coefficients of 0.5 – 0.9, which, for this artificial-data simulation study, can result in one feature being included in a selected set on some data sets and a different but correlated feature being included on others. These results suggest that the selected sets of brain regions on the real HD data set (Table 3) should be interpreted as hypothesis-generating for further clinical studies to verify, rather than as a definite set of brain regions. With that said, it is promising that some of the brain regions that were selected already have strong clinical backing (Section 4.1), such as those in the basal ganglia being associated with TTD.

5 | Discussion

We developed a feature-selection method for censored-covariate regression models that can identify different sets of features—referring to fully-observed covariates—associated with the response variable and the censored covariate. Simulations showed

that these sets can be distinguished when signal is adequate. In the PREDICT-HD application, the method suggested partly distinct candidate brain-region profiles for cognitive ability and time-to-diagnosis, with the application-specific stability analysis showing that individual selected regions should not be interpreted as definitive biomarkers but rather as hypothesis-generating for further clinical studies to verify.

The comparison between the min-CV and 1-SE tuning rules gives a practical way to use the method. We recommend treating the 1-SE rule as a conservative default when the aim is to report a sparse, high-specificity set of features and when the data set contains sufficiently strong signal to support such trimming. However, users should also examine the min-CV rule when scientific prior knowledge suggests weaker signal, when features are highly correlated, or when the 1-SE rule returns empty or implausibly sparse selected sets. In these settings, the min-CV rule can be more useful for identifying candidate features that would otherwise be removed by the 1-SE rule. Therefore, users should choose between the two according to the scientific cost of false positives versus false negatives, and, where possible, features selected by both rules should be interpreted as the most stable, while features selected only by the min-CV rule should be treated as weaker candidates requiring further validation.

The HD application in Section 4 illustrates how our method can be used to compare sets of brain regions associated with cognitive ability and time-to-diagnosis of HD. The selected sets from the min-CV model suggest that these two outcomes may be associated with partly distinct brain-region profiles. The most clinically-interpretable finding is the selection of basal-ganglia regions in the time-to-diagnosis model component, which is consistent with established evidence that the caudate, putamen, and pallidum are central to HD progression. At the same time, we remark that, due to weak signal in the data set, the sets of brain regions selected are not definite; rather, they provide plausible hypotheses about how different brain regions may relate differently to cognitive ability and time-to-diagnosis. The stability analysis in Section 4.2 clarifies the extent to which the selected brain regions should be trusted. Rather than relying on selected brain regions alone, visual diagnostics such as estimated coefficient paths (Figure 2)

can provide additional insight into the strength and stability of regional associations.

Several modelling choices define the present scope of the method and suggest natural extensions. We developed the method for a single, right-censored covariate, which is the setting needed for the HD application and provides a starting point for more general censored-covariate feature selection. The likelihood construction in Section 2.1 also indicates how the approach could be extended to multiple censored covariates or to other censoring mechanisms, such as interval censoring. We modelled the censored-covariate component using a χ^2 distribution because, in the HD application, the censored covariate is a time-to-event quantity. For other applications, a different distribution may be more appropriate; the same general EM framework can be used once the corresponding conditional moments and likelihood terms are specified.

Our method is computationally expensive—it uses an Expectation-Minimisation (EM) algorithm where, in each EM iteration, the expectation step requires numerical integration, and solving the minimisation step requires a further two iterative algorithms (one to minimise J_y and one to minimise J_x). This cost is the tradeoff of being able to use all observations, whilst correctly accounting for the latent part of those with censored x_i , to fit a model. To understand the approximate time required to obtain a model on a data set, we must consider the two tuning parameters, λ_1 and λ_2 , that need to be chosen from a grid of values using K -fold cross-validation. For example, in the simulation study of Section 3, a five-fold cross-validation fit for a single choice of λ_1 and λ_2 on a data set with $n = 1000$ observations, took approximately 30 seconds. Evaluating the full grid of 900 tuning-parameter choices therefore required about 7.5 hours of computation for one data set. To speed up the computation, multiple choices of λ_1 and λ_2 could be evaluated in parallel over multiple computing cores. For the mentioned simulation study which contained $B = 100$ simulated data sets per each of the four simulation settings, we parallelised the model-fitting process over 136 cores on the Gadi supercomputer operated by Australia's National Computational Infrastructure (NCI), allowing us to run a 3000-hour simulation study within 48 hours. To further reduce computational burden, coming up with an information-criterion approach would be a fruitful avenue of further research, as such an approach would avoid having to fit K models per choice of λ_1 and λ_2 .

Future work could consider grouped or structured penalties motivated by the HD application. There is high correlation amongst certain groups or pairs of brain regions, with the selected sets and coefficient paths showing that some left and right brain regions behave similarly. Penalties that group bilateral regions, encode known neuroanatomical pathways, or otherwise share information across related structures could improve stability and interpretability while still allowing the response-variable and censored-covariate model components to select different profiles. Moreover, the HD analysis used study-entry measurements only, which was a simplification that gave one observation per individual and allowed us to assume independence across observations. Building upon our method to include random effects would relax this simplification by using repeated follow-up measurements while accounting for within-individual dependence. More general extensions, including multiple censored covariates, other

censoring mechanisms, alternative censored-covariate distributions, and random effects, would further broaden the range of applications.

Our method makes it possible to not only identify features associated with a response variable while accounting for a censored covariate, but also to identify features associated with that covariate. In the HD application, this meant separating brain-region profiles linked to two scientifically related quantities that could not previously be distinguished within a censored-covariate feature-selection framework. Telling apart what is linked to cognitive ability from time-to-diagnosis matters because cognitive ability, measured by SDMT, is a candidate endpoint in HD clinical trials. Knowing the features behind it can inform which patients to enrol and which measurements to track. More broadly, the method applies to any right-censored covariate of scientific interest, such as time-to-event quantities observed only up to the end of follow-up, and could be adapted to other censoring mechanisms, such as biomarkers observed only above a lower detection limit, allowing important features to be identified to inform downstream clinical decision making.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

PREDICT-HD data are available upon request from the CHDI Foundation, Inc.

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APPENDIX

A | Simplifying the Q-Function

We show how the Q -function in Section 2.2 simplifies into the sum of two separate loss functions, J_y and J_x , and a constant $K^{(k)}$. Recall from the Expectation Step of Section 2.2 that

$$\begin{aligned} Q(\theta; \theta^{(k)}) &= -\frac{1}{n} \sum_{i: \delta_i=1} \{\log f(y_i | x_i, \mathbf{z}_i; \theta) + \log f(x_i | \mathbf{z}_i; \theta)\} \\ &\quad - \frac{1}{n} \sum_{i: \delta_i=0} \left[E \left\{ \log f(y_i | X_i, \mathbf{z}_i; \theta) \mid \mathcal{D}_i; \theta^{(k)} \right\} \right. \\ &\quad \left. + E \left\{ \log f(X_i | \mathbf{z}_i; \theta) \mid \mathcal{D}_i; \theta^{(k)} \right\} \right] \\ &\quad + \frac{\lambda_1}{\sigma^2} \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\gamma}\|_1. \end{aligned}$$

We first simplify the contribution involving $f(y_i | x_i, \mathbf{z}_i; \theta)$. From the normal response-variable model component,

$$\begin{aligned} \log f(y_i | x, \mathbf{z}_i; \theta) &= -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (y_i - \beta_0 - \alpha x - \boldsymbol{\beta}^\top \mathbf{z}_i)^2. \end{aligned}$$

For observations with censored x_i , taking the expectation conditioned on $\mathcal{D}_i$ and $\theta^{(k)}$ gives

$$\begin{aligned} E \left\{ (y_i - \beta_0 - \alpha X_i - \boldsymbol{\beta}^\top \mathbf{z}_i)^2 \mid \mathcal{D}_i; \theta^{(k)} \right\} &= (y_i - \beta_0 - \alpha m_i^{(k)} - \boldsymbol{\beta}^\top \mathbf{z}_i)^2 + \alpha^2 s_i^{2(k)}, \end{aligned}$$

where $m_i^{(k)}$ and $s_i^{(k)}$ are defined in the Expectation Step of Section 2.2. This equality follows by writing the squared term as a quadratic function of X_i and using $V(X_i | \mathcal{D}_i; \theta^{(k)}) = s_i^{2(k)}$. Therefore, after absorbing the constant $-\frac{1}{2} \log(2\pi)$ into $K^{(k)}$, the contribution of the response-variable model component to $Q(\theta; \theta^{(k)})$ is

$$\begin{aligned} J_y(\alpha, \boldsymbol{\beta}, \beta_0, \sigma^2; \theta^{(k)}) &= \frac{1}{2n\sigma^2} \sum_{i: \delta_i=1} (y_i - \beta_0 - \alpha x_i - \boldsymbol{\beta}^\top \mathbf{z}_i)^2 \\ &\quad + \frac{1}{2n\sigma^2} \sum_{i: \delta_i=0} \left[(y_i - \beta_0 - \alpha m_i^{(k)} - \boldsymbol{\beta}^\top \mathbf{z}_i)^2 + \alpha^2 \{s_i^{(k)}\}^2 \right] \\ &\quad + \frac{1}{2} \log \sigma^2 + \frac{\lambda_1}{\sigma^2} \|\boldsymbol{\beta}\|_1. \end{aligned}$$

After reordering the observations so that those with complete x_i appear before those with censored x_i , the function J_y can be written compactly using matrix-vector notation as

$$\begin{aligned} J_y(\alpha, \boldsymbol{\beta}, \beta_0, \sigma^2; \theta^{(k)}) &= \frac{1}{2n\sigma^2} (\|\mathbf{y} - \beta_0 \mathbf{1}_n - \alpha \mathbf{w}^{(k)} - \mathbf{Z}\boldsymbol{\beta}\|_2^2 + \|\alpha \mathbf{s}^{(k)}\|_2^2) \\ &\quad + \frac{1}{2} \log \sigma^2 + \frac{\lambda_1}{\sigma^2} \|\boldsymbol{\beta}\|_1, \end{aligned}$$

where $\mathbf{y}$, $\mathbf{Z}$, $\mathbf{w}^{(k)}$ and $\mathbf{s}^{(k)}$ are defined in the Minimisation Step of Section 2.2.

We next simplify the contribution involving $f(x_i | \mathbf{z}_i; \theta)$. Let $\mu_i = \exp(\gamma_0 + \boldsymbol{\gamma}^\top \mathbf{z}_i)$. The $\chi^2(\mu_i)$ density is

$$f(x | \mathbf{z}_i; \theta) = \frac{1}{2^{\mu_i/2} \Gamma(\mu_i/2)} x^{\mu_i/2-1} \exp\left(-\frac{x}{2}\right), \quad x > 0.$$

Taking the logarithm gives

$$\begin{aligned} \log f(x | \mathbf{z}_i; \theta) &= -\frac{\mu_i}{2} \log 2 - \log \Gamma\left(\frac{\mu_i}{2}\right) + \left(\frac{\mu_i}{2} - 1\right) \log x - \frac{x}{2} \\ &= -\frac{1}{2} (\log 2 - \log x) \mu_i - \log \Gamma\left(\frac{\mu_i}{2}\right) - \log x - \frac{x}{2}. \end{aligned}$$

For observations with complete x_i , substituting $x = x_i$ into the log-density and absorbing the terms that do not depend on γ_0 or $\boldsymbol{\gamma}$ into $K^{(k)}$ gives

$$-\frac{1}{2} (\log 2 - \log x_i) \exp(\gamma_0 + \boldsymbol{\gamma}^\top \mathbf{z}_i) - \log \Gamma\left(\frac{1}{2} \exp(\gamma_0 + \boldsymbol{\gamma}^\top \mathbf{z}_i)\right).$$

For observations with censored x_i , taking the expectation of the log-density conditioned on $\mathcal{D}_i$ and $\theta^{(k)}$ and again absorbing terms

that do not depend on γ_0 or γ into $K^{(k)}$ gives

$$-\frac{1}{2}(\log 2 - \ell_i^{(k)}) \exp(\gamma_0 + \gamma^\top \mathbf{z}_i) - \log \Gamma \left\{ \frac{1}{2} \exp(\gamma_0 + \gamma^\top \mathbf{z}_i) \right\},$$

where $\ell_i^{(k)}$ is defined in the Expectation Step of Section 2.2. Therefore, the contribution of the censored-covariate model component to $Q(\theta; \theta^{(k)})$ is

$$\begin{aligned} J_x(\gamma, \gamma_0; \theta^{(k)}) &= \frac{1}{2n} \sum_{i: \delta_i=1} (\log 2 - \log x_i) \exp(\gamma_0 + \gamma^\top \mathbf{z}_i) \\ &\quad + \frac{1}{2n} \sum_{i: \delta_i=0} (\log 2 - \ell_i^{(k)}) \exp(\gamma_0 + \gamma^\top \mathbf{z}_i) \\ &\quad + \frac{1}{n} \sum_{i=1}^n \log \Gamma \left\{ \frac{1}{2} \exp(\gamma_0 + \gamma^\top \mathbf{z}_i) \right\} + \lambda_2 \|\gamma\|_1. \end{aligned}$$

With scalar functions applied entry-wise and observations re-ordered so that those with complete x_i appear before those with censored x_i , the function J_x can be written compactly using matrix-vector notation as

$$\begin{aligned} J_x(\gamma, \gamma_0; \theta^{(k)}) &= \frac{1}{2n} \mathbf{b}^{(k)\top} \exp(\gamma_0 \mathbf{1}_n + \mathbf{Z}\gamma) \\ &\quad + \frac{1}{n} \mathbf{1}_n^\top \log \Gamma \left\{ \frac{1}{2} \exp(\gamma_0 \mathbf{1}_n + \mathbf{Z}\gamma) \right\} + \lambda_2 \|\gamma\|_1, \end{aligned}$$

where $\mathbf{b}^{(k)}$ is defined in the Minimisation Step of Section 2.2.

Combining the compact expressions for J_y and J_x , the Q -function can be expressed as

$$Q(\theta; \theta^{(k)}) = J_y(\alpha, \beta, \beta_0, \sigma^2; \theta^{(k)}) + J_x(\gamma, \gamma_0; \theta^{(k)}) + K^{(k)},$$

where $K^{(k)}$ collects all terms that do not depend on θ . Since J_y depends only on $[\alpha, \beta, \beta_0, \sigma^2]$ and J_x depends only on $[\gamma, \gamma_0]$, the M-step separates into two minimisation problems.

B | Global Minimisation of J_y

We show how to globally minimise J_y by profiling out σ^2 . Recall from the Minimisation Step of Section 2.2 that

$$\begin{aligned} J_y(\alpha, \beta, \beta_0, \sigma^2; \theta^{(k)}) &= \frac{1}{2n\sigma^2} (\|\mathbf{y} - \beta_0 \mathbf{1}_n - \alpha \mathbf{w}^{(k)} - \mathbf{Z}\beta\|_2^2 + \|\alpha \mathbf{s}^{(k)}\|_2^2) \\ &\quad + \frac{1}{2} \log \sigma^2 + \frac{\lambda_1}{\sigma^2} \|\beta\|_1. \end{aligned}$$

We begin by univariate minimising J_y over σ^2 , keeping the other parameters fixed. Let

$$A(\alpha, \beta, \beta_0) = \frac{1}{2n} (\|\mathbf{y} - \beta_0 \mathbf{1}_n - \alpha \mathbf{w}^{(k)} - \mathbf{Z}\beta\|_2^2 + \|\alpha \mathbf{s}^{(k)}\|_2^2) + \lambda_1 \|\beta\|_1.$$

Then the response-variable loss can be written as

$$J_y(\alpha, \beta, \beta_0, \sigma^2; \theta^{(k)}) = \frac{A(\alpha, \beta, \beta_0)}{\sigma^2} + \frac{1}{2} \log \sigma^2.$$

For fixed $[\alpha, \beta, \beta_0]$, differentiating with respect to σ^2 gives

$$\frac{\partial J_y}{\partial \sigma^2} = -\frac{A(\alpha, \beta, \beta_0)}{(\sigma^2)^2} + \frac{1}{2\sigma^2}.$$

Thus, as long as $A(\alpha, \beta, \beta_0) > 0$, the unique minimiser over σ^2 for those fixed parameters is

$$\sigma^2 = 2A(\alpha, \beta, \beta_0).$$

Substituting this value back into J_y yields the profiled loss function

$$\frac{1}{2} + \frac{1}{2} \log \{2A(\alpha, \beta, \beta_0)\},$$

which is a strictly increasing function of $A(\alpha, \beta, \beta_0)$. Therefore, jointly globally minimising J_y over $[\alpha, \beta, \beta_0, \sigma^2]$ is equivalent to first globally minimising $A(\alpha, \beta, \beta_0)$ over $[\alpha, \beta, \beta_0]$, and then setting σ^2 equal to twice the minimum value of A .

Global minimisation of $A(\alpha, \beta, \beta_0)$ can be achieved using a cyclic coordinate descent algorithm since it is convex. Denoting the minimiser by $[\alpha^{(k+1)}, \beta^{(k+1)}, \beta_0^{(k+1)}]$, the corresponding variance update is

$$\begin{aligned} \sigma^{2(k+1)} &= 2A(\alpha^{(k+1)}, \beta^{(k+1)}, \beta_0^{(k+1)}) \\ &= \frac{1}{n} (\|\mathbf{y} - \beta_0^{(k+1)} \mathbf{1}_n - \alpha^{(k+1)} \mathbf{w}^{(k)} - \mathbf{Z}\beta^{(k+1)}\|_2^2 + \|\alpha^{(k+1)} \mathbf{s}^{(k)}\|_2^2) \\ &\quad + 2\lambda_1 \|\beta^{(k+1)}\|_1. \end{aligned}$$

Thus, J_y is minimised globally. This global optimality is precisely where the scaling of the L_1 penalty by $1/\sigma^2$ is used: it allows the variance to be profiled out and leaves a convex penalised least-squares problem in $[\alpha, \beta, \beta_0]$.

C | Standardisation of Data and Back-Transformation of Parameter Estimates

Before fitting the model, we standardise the response variable y_i and features $\mathbf{z}_i$ for each observation $i = 1, \dots, n$. We denote the standardised response variable and features as

$$\tilde{y}_i = \frac{y_i - \bar{y}}{s_y}, \quad \tilde{\mathbf{z}}_i = \mathbf{S}_z^{-1}(\mathbf{z}_i - \bar{\mathbf{z}}), \quad (\text{C1})$$

where $\bar{y}$ and s_y are the sample mean and sample standard deviation of the response variable, $\bar{\mathbf{z}}$ is the vector of sample means of the features, and $\mathbf{S}_z = \text{diag}(s_{z_1}, \dots, s_{z_p})$ is the diagonal matrix of sample standard deviations of the features.

Let $\tilde{\alpha}, \tilde{\beta}, \tilde{\beta}_0, \tilde{\sigma}^2, \tilde{\gamma}$, and $\tilde{\gamma}_0$ denote the parameter estimates obtained after fitting the model on the standardised data. The response-variable model component on the standardised scale is

$$\tilde{y}_i = \tilde{\beta}_0 + \tilde{\alpha} x_i + \tilde{\beta}^\top \tilde{\mathbf{z}}_i + \tilde{\epsilon}_i, \quad \tilde{\epsilon}_i \sim N(0, \tilde{\sigma}^2), \quad (\text{C2})$$

and the censored-covariate model component on the standardised scale is

$$\tilde{\mu}_i = \exp(\tilde{\gamma}_0 + \tilde{\gamma}^\top \tilde{\mathbf{z}}_i). \quad (\text{C3})$$

We show how to back-transform the parameter estimates on the standardised scale to the scale of the original data so as to obtain the original-scale parameter estimates $\hat{\alpha}, \hat{\beta}, \hat{\beta}_0, \hat{\sigma}^2, \hat{\gamma}$, and $\hat{\gamma}_0$. The original-scale parameter estimates are the estimates reported by

our method. Using (C1), we have that

$$\begin{aligned}\tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{z}}_i &= \tilde{\boldsymbol{\beta}}^\top \mathbf{S}_z^{-1}(\mathbf{z}_i - \bar{\mathbf{z}}) \\ &= \tilde{\boldsymbol{\beta}}^\top \mathbf{S}_z^{-1} \mathbf{z}_i - \tilde{\boldsymbol{\beta}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}},\end{aligned}$$

which can be plugged into (C2) to obtain

$$y_i = s_y \tilde{\beta}_0 - s_y \tilde{\boldsymbol{\beta}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}} + \bar{y} + s_y \tilde{\alpha} x_i + s_y \tilde{\boldsymbol{\beta}}^\top \mathbf{S}_z^{-1} \mathbf{z}_i + s_y \tilde{\varepsilon}_i,$$

implying that the original-scale parameter estimates are

$$\begin{aligned}\hat{\alpha} &= s_y \tilde{\alpha}, \\ \hat{\boldsymbol{\beta}}^\top &= s_y \tilde{\boldsymbol{\beta}}^\top \mathbf{S}_z^{-1} \\ \hat{\beta}_0 &= s_y \tilde{\beta}_0 - s_y \tilde{\boldsymbol{\beta}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}} + \bar{y} \\ \hat{\sigma}^2 &= s_y \tilde{\sigma}^2.\end{aligned}$$

Then, again using (C1), we have that

$$\begin{aligned}\tilde{\boldsymbol{\gamma}}^\top \tilde{\mathbf{z}}_i &= \tilde{\boldsymbol{\gamma}}^\top \mathbf{S}_z^{-1}(\mathbf{z}_i - \bar{\mathbf{z}}) \\ &= \tilde{\boldsymbol{\gamma}}^\top \mathbf{S}_z^{-1} \mathbf{z}_i - \tilde{\boldsymbol{\gamma}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}},\end{aligned}$$

which can be plugged into (C3) to obtain

$$\mu_i = \exp\left(\tilde{\gamma}_0 - \tilde{\boldsymbol{\gamma}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}} + \tilde{\boldsymbol{\gamma}}^\top \mathbf{S}_z^{-1} \mathbf{z}_i\right)$$

implying that the original-scale estimates are

$$\begin{aligned}\hat{\gamma}_0 &= \tilde{\gamma}_0 - \tilde{\boldsymbol{\gamma}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}} \\ \hat{\boldsymbol{\gamma}}^\top &= \tilde{\boldsymbol{\gamma}}^\top \mathbf{S}_z^{-1}.\end{aligned}$$

D | Right-Censoring Rate in Simulation Studies

In the simulation studies, we generate $X_i \mid \mathbf{Z}_i = \mathbf{z}_i \sim \chi^2(\mu_i)$ with $\mu_i = \exp(\gamma_0 + \boldsymbol{\gamma}^\top \mathbf{z}_i)$ and independently generate $C_i \mid \mathbf{Z}_i = \mathbf{z}_i \sim \chi^2(\nu_i)$. The parameter ν_i is chosen so that the conditional probability of right-censoring is equal to a target value r_{cens} :

$$P(C_i < X_i \mid \mathbf{Z}_i = \mathbf{z}_i) = r_{\text{cens}}.$$

Both X_i and C_i , being chi-square random variables, are more generally gamma random variables with common scale parameter 2:

$$\begin{aligned}X_i \mid \mathbf{Z}_i = \mathbf{z}_i &\sim \text{Gamma}\left(\frac{\mu_i}{2}, 2\right), \\ C_i \mid \mathbf{Z}_i = \mathbf{z}_i &\sim \text{Gamma}\left(\frac{\nu_i}{2}, 2\right).\end{aligned}$$

It follows that

$$\frac{X_i}{X_i + C_i} \mid \mathbf{Z}_i = \mathbf{z}_i \sim \text{Beta}\left(\frac{\mu_i}{2}, \frac{\nu_i}{2}\right).$$

Therefore,

$$\begin{aligned}P(C_i < X_i \mid \mathbf{Z}_i = \mathbf{z}_i) &= P\left(\frac{X_i}{X_i + C_i} > \frac{1}{2} \mid \mathbf{Z}_i = \mathbf{z}_i\right) \\ &= 1 - F_{\text{Beta}}\left(\frac{1}{2}; \frac{\mu_i}{2}, \frac{\nu_i}{2}\right),\end{aligned}$$

where $F_{\text{Beta}}(\cdot; a, b)$ is the cumulative distribution function of a Beta(a, b) random variable. Hence, ν_i is the solution to the one-dimensional equation

$$1 - F_{\text{Beta}}\left(\frac{1}{2}; \frac{\mu_i}{2}, \frac{\nu_i}{2}\right) - r_{\text{cens}} = 0.$$

The left-hand side is monotone decreasing in ν_i , so the solution can be found using a standard root-finding method such as one of those implemented in `uniroot` in R. In the special case $r_{\text{cens}} = 50\%$, symmetry gives $\nu_i = \mu_i$.